

MIAO LI, PH.D.

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Senior Scientist operating at the interface of AI-enabled software engineering, computational biology, and immunology, with a track record of building AI-powered platforms, deep learning models, and multi-omics/immune-repertoire pipelines that translate single-cell and NGS data into translational insights supporting TCR discovery and cell therapy discovery. Grounded in hands-on CAR/TCR-T cell engineering — bridging computational and experimental discovery — with strong cross-functional project leadership across multidisciplinary programs.

PROFESSIONAL EXPERIENCE

AstraZeneca | Senior Scientist | Gaithersburg, MD

2023.11 – Present

- **AI Architecture & Enablement:** Architect and lead a reusable, AI-powered agent that automates NGS bioinformatics workflows — translating natural-language chat into SLURM-orchestrated pipelines on a shared HPC cluster and giving wet-lab scientists self-serve access to TCR/antibody discovery analysis without coding — removing the bioinformatics support bottleneck and accelerating the sequencing-to-insight loop across the BE discovery teams.
- **Bioinformatics:** Build, lead, and maintain in-house immune-repertoire NGS and bulk/single-cell RNA-seq pipelines with QC and reproducibility standards, streamlining end-to-end TCR-seq workflows from raw reads to annotated full-length TCR sequences and accelerating discovery of antigen-reactive clones in support of Oncology and Respiratory & Immunology (R&I) research.
- **Data Science:** Train a customized deep learning-based TCR classifier on public TCR-repertoire data spanning four disease states to predict antigen reactivity, translating computational predictions into experimentally validated candidates. Applied to a ~7,000-clone immune-repertoire NGS screening panel, predictions validated at a 35% hit rate against MART-1 (an overexpressed melanoma peptide), recovering functional clones as rare as 0.0002% frequency.
- **TCR Discovery:** Conduct antigen-specific TCR screening spanning antigen-reactive T-cell expansion, lentiviral production/transduction, TCR-library electroporation, and functional co-culture characterization with peptide-pulsed APCs to support reactive-TCR discovery in oncology and autoimmune indications. Lead end-to-end single-cell RNA/TCR-seq multi-omic analysis supporting biomarker and endotype discovery in Graves' disease, integrating wet-lab screening readouts with gene-expression and immune-repertoire cross-dataset mining; prioritized ~70 antigen-reactive clonotypes from a raw repertoire of ~3,000 NGS-derived clones and identified 18 experimentally and computationally validated true hits, informing target selection for TCR-blockade therapeutic design.
- **Cell Therapy Discovery:** Developed a chemically defined, stem cell-derived "off-the-shelf" CAR-T differentiation protocol achieving >100-million-fold functional T-cell expansion, outperforming commercial benchmarks; characterized stem-cell-to-T-cell developmental trajectories via scRNA-seq and evaluated gene-engineering strategies for forward-programmed differentiation using virtual perturbation modeling. Partner cross-functionally to build bulk/scRNA-seq and TCR-seq bioinformatics workflows supporting Treg cell-therapy discovery, identifying differentially expressed genes and enriched pathways across Treg subpopulations, contributing to one peer-reviewed publication.

University of California, Los Angeles | Postdoctoral Fellow | Los Angeles, CA

2022.01 – 2023.10

- **Cell Therapy Bioinformatics:** Led comprehensive scRNA-seq/TCR-seq analyses profiling gene-expression signatures of conventional CAR-T and stem cell-derived allogeneic CAR-NKT cells against solid tumors, revealing insights into in vivo persistence, tumor trafficking, and intratumoral gene expression, and uncovering novel CAR-T/immune-blockade combination therapy strategies.

Indiana University – Bloomington | Postdoctoral Fellow | Bloomington, IN

2018.09 – 2022.01

- **Computational Imaging:** Applied machine learning-guided image analysis to quantitative super-resolution microscopy data of immune receptors in macrophages, revealing previously unseen relationships between innate immune activation and receptor spatial organization at the nanoscale.

PROJECTS

▪ **NGS Intelligent Workflow Assistant – AI Agent for Bioinformatic Pipelines (2026.03 – 2026.08)**

- To make NGS bioinformatics analysis self-serve for non-technical bench scientists in BE — removing the command-line tooling, manual scripting, and multi-step HPC setup that blocked adoption — I built an AI agent that automates antibody/TCR sequencing pipelines on a shared HPC/SLURM cluster through natural-language chat.
- I built a full-stack conversational agent (Claude Agent SDK; TypeScript/React/Node; MCP) that elicits NGS workflow parameters through chat and turns them into SLURM job submissions across 10x/Smart-seq TCR sequencing, antibody scFv/Fab/VHH discovery (Illumina and PacBio NGS), and clonal repertoire analysis — tracking job lifecycle and running interactive analysis and plotting in a per-user sandbox. It's safeguarded by human-in-the-loop approval, per-user identity isolation, and audit logging, delivered through a spec- and test-driven process (PRD → spec → plan → tasks, independent code review), and validated against real SLURM runs.
- The result: zero-CLI, no-coding job submission in under five minutes, reproducible and fully traceable runs by default, and an extensible, reusable architecture designed to onboard new bioinformatics analyses as skills — lowering the bioinformatics expertise barrier and accelerating the sequencing-to-insight loop for the entire TCR/antibody discovery group.

- **TCR Pathology Classifier – Deep Learning modeling for Disease-reactive TCR Discovery (2025.09 – 2026.03)**
 - To compress the months-long, wet-lab-heavy search for ultra-rare disease-reactive TCRs, I built a protein language model (PLM)-based deep learning classifier that prioritizes disease-reactive TCR clonotypes directly from sequence, enabling early down-selection of candidates without the ~8-week experimental enrichment process.
 - I trained a customized ESM2-based deep learning framework with a cross-attention architecture over paired TRA/TRB CDR3 sequences and V/J gene features on the McPAS-TCR database (covering four disease states), fine-tuned end-to-end in PyTorch, and deployed it as a Flask web application for inference. I validated the model in a proof-of-concept study on a 6,737-clone healthy-donor raw NGS repertoire pulsed with the MART-1 melanoma antigen.
 - The classifier prioritized ~7,000 raw clones down to 572 melanoma-associated candidates (a ~10x reduction in search space); of 31 clones advanced to functional testing, 11 validated as MART-1-reactive (a 35% hit rate), recovering true hits as rare as 0.0002% frequency and compressing the raw-repertoire-to-shortlist timeline from ~8 weeks to 10 days.

EDUCATION

University of Munich | Munich, Germany | 2013 – 2017

Doctor of Natural Science (Ph.D.) in Biophysics

The University of Nottingham | Nottingham, UK | 2011 – 2012

Master of Research (M.Res.) in Pharmacy

Wuhan University of Technology | Wuhan, China | 2006 – 2011

Bachelor of Engineering in Pharmaceutical Science

SELECTED PUBLICATIONS

- Y-R. Li, **M. Li***, Y. Chen, et al. Spatiotemporal profiling reveals distinct dynamics and checkpoint regulations of CAR-T and CAR-NKT cells against solid tumors, **Signal Transduction and Targeted Therapy**, 2026 Mar 11, 92. (*co-first authors)
- F. Yang, A. Yanes, **M. Li**, P. Heizer, et al. Differential regulation of Treg stability in human naïve and effector Treg subsets by TGFβ-signaling via ARKADIA-SKI axis, **Frontiers in Immunology**, 2025 Sep 9, 16.
- Y-R. Li, Z. Li, Y. Zhu, **M. Li**, Y. Chen, et al. Overcoming ovarian cancer resistance and evasion to CAR-T cell therapy by harnessing allogeneic CAR-NKT cells. **Med** 2025 Aug 12, (6) 10.
- B. Li, J. Brown, **M. Li**, E. Zhu, et al. Serotonin transporter inhibits antitumor immunity through regulating the intratumoral serotonin axis. **Cell**. 2025 Jul 10, 188(14):3823-3842.e21.
- Y-R. Li, Y. Zhou, J. Yu, **M. Li**, et al. Generation of allogeneic CAR-NKT cells from hematopoietic stem and progenitor cells using a clinically guided culture method. **Nature Biotechnology**. 2025 Mar, 43(3):329-344.
- Y-R. Li, Z. Dunn, **M. Li**, et al. Advancing cell-based cancer immunotherapy through stem cell engineering. **Cell Stem Cell**, 2023, 30.
- Y. Zhou*, **M. Li***, K. Zhou, J. Brown, T. Tsao, X. Cen, T. Husman, A. Bajpai, Z. Dunn, L. Yang. Engineering-induced pluripotent stem cells for cancer immunotherapy. **Cancers**, 2022, 14 (9), 2266. (*co-first authors)
- **M. Li**, C. Vultorius, M. Bethi, Y. Yu. Spatial organization of Dectin-1 and TLR2 during synergistic crosstalk revealed by super-resolution imaging. **The Journal of Physical Chemistry B**, 2022, 126, 31, 5781–5792.
- **M. Li**, S. Lee, M. Zahedian, C. Ding, J. Yan, Y. Yu. Immobile ligands enhance FcγR-TLR2/1 crosstalk by promoting interface overlap of receptor clusters. **Biophysical Journal**, 2022, 6, 966-976.
- **M. Li**, Y. Yu. Clustering of Innate Immune Receptors and Immune Regulation. **Journal of Cell Sciences**, 2021, 134 (4).
- **M. Li**, H. Wang, W. Li, X. G. Xu, Y. Yu, Macrophage activation on “phagocytic synapse” arrays: Spacing of nanoclustered ligands directs TLR1/2 signaling with an intrinsic limit. **Science Advances**, 2020, 6, eabc8482.
- Y.-q. Yu, **M. Li**, Y. Yu. Tracking single molecules in biomembranes: Is seeing always believing? **ACS Nano**, 2019, 13, 10860-10868.
- **M. Li**, T. Lohmüller, J. Feldmann. Optical injection of gold nanoparticles into living cells. **Nano Letters**, 2015, 15 (1), 770–775.

TECHNICAL SKILL SET

Computational Skills:

- **Programming:** Python, R, Shell scripting, UNIX/Linux, HPC clusters, Git, vibe coding with SDD (spec-driven development)
- **AI/ML/Data Science:** Scikit-learn for classical ML (Linear/Logistic Regression, Random Forest, XGBoost, K-means, DBSCAN, PCA), PyTorch for deep learning architectures (CNN, Transformer, Attention, encode- decoders); LLM application development (Claude Agent SDK, MCP, LangChain/LangGraph, RAG); data visualization (Matplotlib, Seaborn, ggplot2, Tidyverse)
- **Bioinformatics:** Bulk and single-cell RNA sequencing, scTCR-seq, immune-repertoire analysis (MiXCR), CellRanger, Scanpy/Seurat (full workflow from QC and alignment through integration, clustering, annotation, differential expression, pathway analysis, and pseudotime/RNA-velocity; cell–cell communication; gene-perturbation prediction; large-scale multi-omics data management.

Wet-Lab Techniques:

- **Cellular Immunology & CAR-T Engineering:** Primary cell culture and expansion (HSCs, iPSCs, PBMCs, CAR-T, NK, iNKT, macrophages); 2D/3D iPSC differentiation into effector T cells (e.g., CD8+ CAR-T); multi-color flow cytometry (T-cell phenotyping); ELISA (cytokine profiling); cytotoxicity/viability assays (xCELLigence, plate-reader killing assays)
- **Molecular Biology & Gene Editing:** CRISPR/Cas9-mediated gene knock-in and editing, lentivirus production, packaging, and transduction (CAR-T engineering), molecular cloning (Gibson assembly), PCR, genomic DNA extraction, Western blot.